ME 494

Maintenance Engineering

Course Objectives

1. Understanding and estimate the cost components of maintenance
2. Effect of engineering activities on the environment
3. Understanding of different types of maintenance practices that influences machine readiness
4. Finding logical solution to machine problems
5. Safety awareness through the knowledge of the various hazards at the work places.

Course Objectives Cont'd

6. Identify the human errors in maintenance engineering
7. Communicate to technician how components must be welded together
8. When and how to request for oil analysis and utilize the oil report to decode machine status
9. Write report on maintenance activities to management.
10. Skills of a maintenance engineer be known and used in job seeking.

MAINTENANCE ENGINEERING

Maintenance:

Is the all appropriate actions for retaining an item/part/equipment in, or restoring it to, a given condition.

In short the upkeep of property or equipment

MAINTENANCE ENGINEERING

Engineering

Is the application of science and mathematics by which the properties of matter and the source of energy in nature are made useful to people.

Maintenance Engineering

Maintainability

Is the probability that a failed item/equipment/ machine will be restored to a required working condition

Maintenance Engineering

Is the profession and discipline which apply engineering concepts to optimize equipment, procedures, department budget, and environment to achieve better reliability, availability and maintainability of equipment.

Whoever is engaged in such activities is referred to as Maintenance Engineer.

Maintenance Engineering Departments

- Equipment is found in every sector of the economy and must operate when needed
- Anytime the equipment does what it needs to do when it needs to do it, plant output and profitability is maximized
- They must be manned to correct design flaws, material imperfection and life in order to achieve the maximum equipment operations.
- It is the goal of every establishment to carry out flawless execution of duties to please its share holders by providing very good returns on their investments

The Maintenance Engineer

- The equipment requires regular services as demanded by manufacturers
- These departments need manpower to function hence the preparedness to occupy such positions
- the maintenance engineer is needed at all times!
- GOOD Maintenance Engineers to take up such ever increasing challenging jobs

Industries which employs maintenance engineers

- Engineering construction
- Process industries including pharmaceuticals, food and cosmetics
- Aerospace and automotive
- Railways
- Electronics and IT engineering
- Manufacturing
- Armed forces
- Government agencies
- Engineering consultancies
- Medical engineering
- Energy utilities including nuclear
- Oil, gas and petro-chemical
- Research agencies
- Sea transportation

Responsibility of maintenance engineering department or maintenance engineer

- Planning and repairing equipment/facilities to acceptable standards in accordance with best practices.
- Performing preventive maintenance; more specifically, developing and implementing a regularly scheduled work program for the purpose of maintaining satisfactory equipment/facility operation as well as preventing major problems
- Preparing realistic budgets that detail maintenance personnel and material needs i.e. estimation of maintenance cost and evaluation of alternatives
- Managing inventory to ensure that parts/materials necessary to conduct maintenance tasks are readily available i.e. forecasting of spare parts
- Keeping records on equipment, services, etc.
- Analysis of repetitive equipment failures

- Developing effective approaches to monitor the activities of maintenance staff i.e. reviewing personnel transfer to and from the maintenance department
- Developing effective techniques for keeping operations personnel, upper-level management, and other concerned groups aware of maintenance activities
- Training maintenance staff and other concerned individuals to improve their skills and perform effectively i.e. assessing required maintenance tools and skills needed for efficient maintenance of equipment
- Reviewing plans for new facilities, installation of new equipment, etc. i.e. assessing the needs for equipment replacements and establish replacement programs when due
- Implementing methods to improve workplace safety and developing safety education-related programs for maintenance staff i.e. assessing and reporting safety hazards associated with maintenance of equipment

- Developing contract specifications and inspecting work performed by contractors to ensure compliance with contractual requirements
- Adhering to strict environmental standards with respect to waste disposal
- Assure optimization of the maintenance department structure.
- Assisting research and development (R&D) with the investigation and new product development.

Goals of the Maintenance Engineering Department

1. Healthy operational equipment
2. Safety
3. Environment
4. Cost
5. Right tools
6. Right personnel/staff
7. Knowledge of the appropriate technology

Maintenance Engineering Skills

Self:

With the type of maintenance exercises, the engineer works in time constrain limits. During these times, the work place is tensed and pressurized and the engineer must be very well composed to execute the work.

The engineer must exhibit self confidence, excellent logic based approach to problem solving and unyielding effort in achieving successful completion of exercise.

Technical:

Once a while the engineer will design a component to replace a time-constraint exercise.

The engineer must be knowledgeable in welding practices, wear rate, temperature and pressure related issues, data collection, data analysis, corrosion, fatigue, and cracks.

The engineer must understand selection of bearing, types of drive belts, and chains.

The engineer must be abreast with different types of welding practices.

The engineer must read blue prints easily.

All the above reinforce quality mechanical engineering training.

In short, being a maintenance engineer require the use of all the courses read in mechanical engineering.

Legal:

Maintenance engineer liaise with the legal department about claims under warranty by providing technical information as to why the claims hold.

The engineer must adhere to manufacturers' demands as to how a machine must be operated and serviced in order to qualify for warranty.

The engineer needs to understand all contract documents concerning the machines in question.

Social:

The engineer be able to tolerate and resolve both social and technical issues with both upper and lower management teams.

The engineer should sound hope to stressed workers (engineers, technicians, craftsmen, etc) to achieve timely successful exercise.

Financial Adviser

The engineer advises the company on maintenance cost, machine profitability of liability and finally the disposal and replacement of machines/equipment.

The engineer notifies the upper management about the size of workforce.

Environment and Safety

The engineer communicates and reinforces environmental requirement and safety equipment to both upper and lower management teams.

Structure of Maintenance Engineering Department

Maintenance engineering department may or may not report to the production department, depending on the size of the establishment/organization.

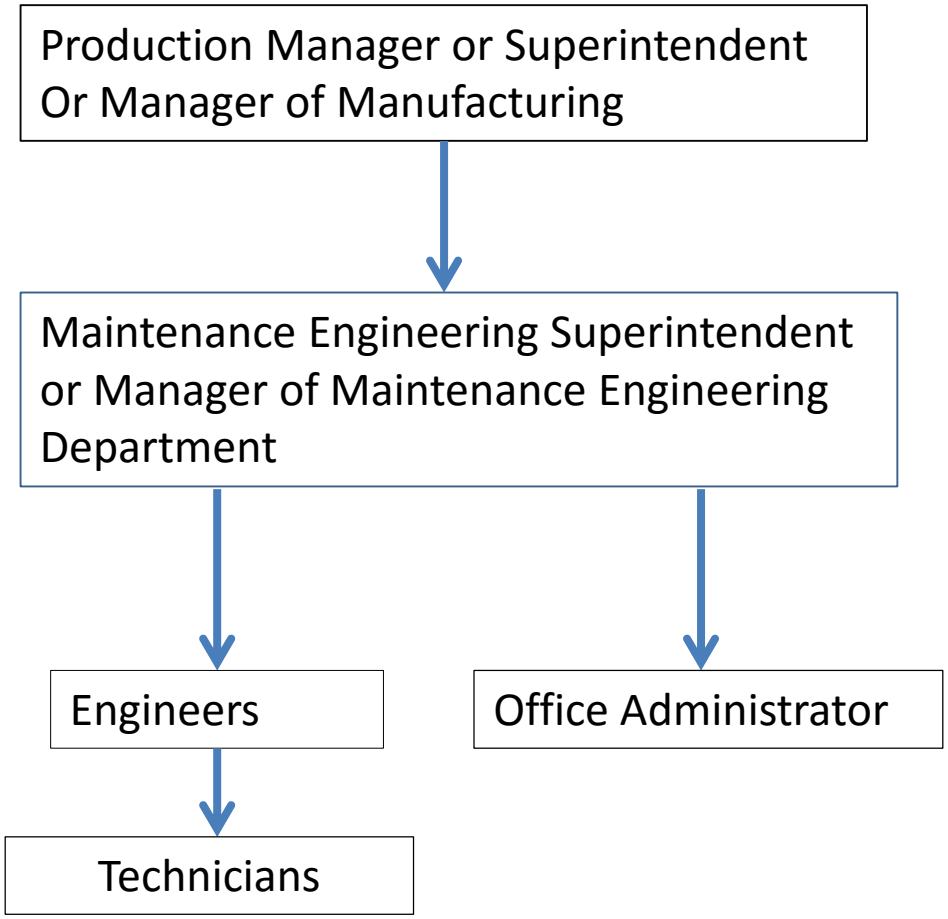
Bigger organizations will separate these departments and the two departments will report to a higher manager.

In some smaller firms maintenance engineering department reports to production department in the organizational structure .

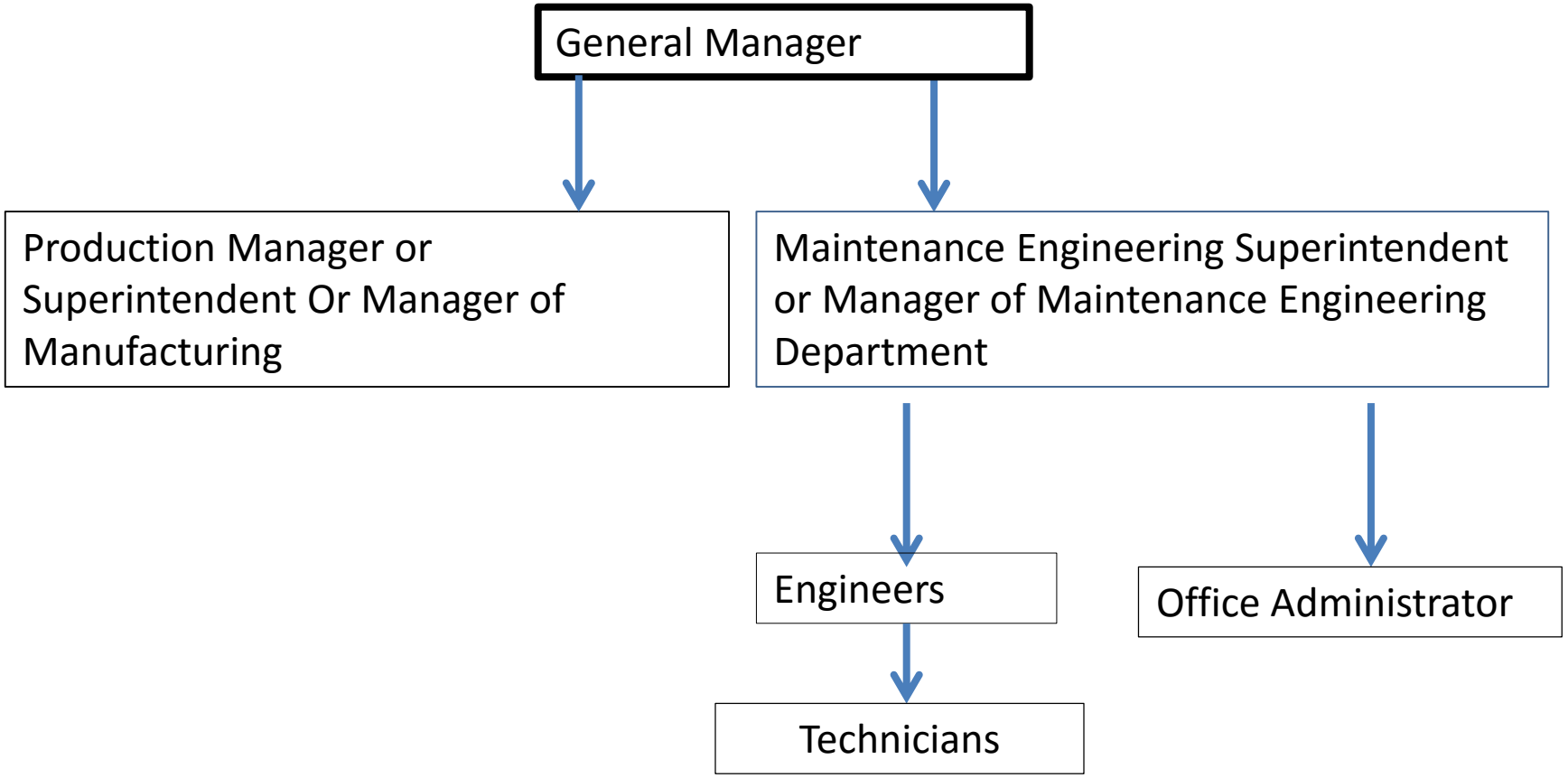
Whatever responsibilities are assigned to the maintenance engineering department, it is important that they be clearly defined and that the limits of authority and established responsibilities agreed upon by all concerned.

The production and maintenance engineering department work closely in making the fortunes of the organization a reality.

Maintenance Department Reporting to Production Manager



Maintenance Department and Production Department at Par and Report to General Manager



Question:

1. What is the difference between maintainability and maintenance?

2. Discuss the need for maintenance.

3. Why should the maintenance engineering and production departments be autonomous?

4. Can maintenance engineering department be part of the production department, how?

5. In an interview, a candidate was asked why she must be employed. What can she say?

6. A new CEO upon assuming office, invites a young engineer to brief him of the activities that maintenance department engages in. What would the engineer write?

ELEMENTS OF EFFECTIVE MAINTENANCE MANAGEMENT

There are many elements of effective maintenance management whose effectiveness is the key to the overall success of the maintenance activity

MAINTENANCE POLICY	PREVENTIVE AND CORRECTIVE MAINTENANCE
MATERIAL CONTROL	JOB PLANNING AND SCHEDULING
WORK ORDER SYSTEM	BACKLOG CONTROL AND PRIORITY SYSTEM
EQUIPMENT RECORDS	PERFORMANCE MEASUREMENT

MAINTENANCE POLICY

- A maintenance policy is one of the most important elements of effective maintenance management.
- It is essential for continuity of operations and a clear understanding of the maintenance management program, regardless of the size of a maintenance organization.
- Usually, maintenance organizations have manuals containing items such as policies, programs, objectives, responsibilities, and authorities for all levels of supervision, reporting requirements, useful methods and techniques, and performance measurement indices.
- Lacking such documentation, i.e., a policy manual, a policy document must be developed containing all essential policy information.

MATERIAL CONTROL

- Experience indicates that, on average, material costs account for approximately 30 to 40% of total direct maintenance costs.
- Efficient utilization of personnel depends largely on effectiveness in material coordination.
- Material problems can lead to false starts, excess travel time, delays, unmet due dates, etc.
- Steps such as job planning, coordinating with purchasing, coordinating with stores, coordination of issuance of materials, and reviewing the completed job can help reduce material related problems.
- Deciding whether to keep spares in storage is one of the most important problems of material control.

WORK ORDER SYSTEM

- A work order authorizes and directs an individual or a group to perform a given task.
- A well-defined work order system should cover all the maintenance jobs requested and accomplished, whether repetitive or one-time jobs.
- The work order system is useful for management in controlling costs and evaluating job performance.
- Although the type and size of the work order can vary from one maintenance organization to another,

A work order should at least contain information such as:

- requested and planned completion dates
- work description and its reasons
- planned start date
- labour and material costs
- item or items to be affected
- work category (preventive maintenance, repair, installation, etc.)
- appropriate approval signatures

EQUIPMENT RECORDS

- Equipment records play a critical role in effectiveness and efficiency of the maintenance organization.
- Usually, equipment records are grouped under four classifications:
  - maintenance work performed
  - maintenance cost
  - inventory
  - files
- The maintenance work performed category contains chronological documentation of all repairs and preventive maintenance (PM) performed during the item's service life to date.

- The maintenance cost category contains historical profiles and accumulations of labour and material costs by item.

- **Usually, information on inventory is provided by the stores or accounting department**. The inventory category contains information such as property number, size and type, procurement cost, date manufactured or acquired, manufacturer, and location of the equipment/item.

- The files category includes operating and service manuals, warranties, drawings, and so on.

Equipment records are useful when:

- procuring new items/equipment to determine operating performance trends
- troubleshooting breakdowns
- making replacement or modification decisions
- investigating incidents
- identifying areas of concern
- performing reliability and maintainability studies
- conducting life cycle cost and design studies.

PREVENTIVE AND CORRECTIVE MAINTENANCE

- The basic purpose of performing PM is to keep facility/equipment in satisfactory condition through inspection and correction of early-stage deficiencies.

- Three principle factors shape the requirement and scope of the PM effort:
  - process reliability
  - economics
  - standards compliance.

- A major proportion of a maintenance organization's effort is spent on corrective maintenance (CM).

- Thus, CM is an important factor in the effectiveness of maintenance organization.

JOB PLANNING AND SCHEDULING

- Job planning is an essential element of the effective maintenance management.
- A number of tasks may have to be performed prior to commencement of a maintenance job; for example,
  - procurement of parts, tools, and materials, coordination and delivery of parts, tools, and materials, identification of methods and sequencing, coordination with other departments, and securing safety permits.

- Although the degree of planning required may vary with the craft involved and methods used, past experience indicates that on average one planner is required for every twenty craftpersons.

- Strictly speaking, formal planning should cover 100% of the maintenance workload but emergency jobs and small, straightforward work assignments are performed in a less formal environment.
- Thus, in most maintenance organizations 80 to 85% planning coverage is attainable.
- Maintenance scheduling is as important as job planning.
- Schedule effectiveness is based on the reliability of the planning function.
- For large jobs, in particular those requiring multi-craft coordination, serious consideration must be given to using methods such as Program Evaluation and Review Technique (PERT) and Critical Path Method (CPM) to assure effective overall control.

BACKLOG CONTROL AND PRIORITY SYSTEM

- The amount of backlog within a maintenance organization is one of the determining factors of maintenance management effectiveness.
- Identification of backlogs is important to balance manpower and workload requirements.
- decisions concerning overtime, hiring, subcontracting, shop assignments, etc., are largely based on backlog information.
- Management makes use of various indices to make backlog related decisions.
- The determination of job priority in a maintenance organization is necessary since it is not possible to start every job the day it is requested.

In assigning job priorities, it is important to consider factors such as

- importance of the item or system
- the type of maintenance
- required due dates
- the length of time the job awaiting scheduling will take.

PERFORMANCE MEASUREMENT

- Successful maintenance organizations regularly measure their performance through various means.

Performance analyses

- contribute to maintenance department efficiency
- Essential to revealing the downtime of equipment
- peculiarities in operational behaviour of the concerned organization
- developing plans for future maintenance

MAINTENANCE MANAGEMENT CONTROL INDICES

- Management employs various approaches to measure effectiveness of the maintenance function.
- Often it uses indices to manage and control maintenance.
- These indices show trends by using past data as a reference point.
- Usually, a maintenance organization employs various indices to measure maintenance effectiveness, as there is no single index that can accurately reflect the overall performance of the maintenance activity.

- The main objective of these indices is to encourage maintenance management to improve on past performance.
- The broad indices indicate the overall performance of the organization with respect to maintenance
- The specific indices indicate the performance in particular areas of the maintenance function.
- The values of all these indices are plotted periodically to show trends.

BROAD INDICATORS

This section presents three such indicators.

Index I

This is defined by $$I_1 = \frac{\mathbf{TMC}}{\mathbf{TS}} \qquad \mathbf{(2.1)}$$

where

TMC = total maintenance cost,

TS = total sales,

I_1 = index parameter.

Past experience indicates that average expenditure for maintenance for all industry was around 5% of sales.

However, there was a wide variation among industries.

For example, the average values of I1 for steel and chemical industries were 12.8 and 6.8%, respectively.

Index II

This is expressed by

$$I_2 = \frac{\text{TMC}}{\text{TO}} \qquad (2.2)$$

where

TO = total output expressed in gallons, tons, megawatts, etc.,

I_2 = index parameter.

This index relates the total maintenance cost to the total output by the organization.

Index III

This is defined as follows:

$$I_3 = \frac{\mathrm{TMC}}{\mathrm{TIPE}} \qquad (2.3)$$

where

I_3 = index parameter,

TIPE = total investment in plant and equipment.

This index relates the total maintenance cost to the total investment in plant and equipment. The approximate average figures for I_3 in the steel and chemical industries are 8.6 and 3.8%, respectively.

SPECIFIC INDICATORS

- Index IV

This is a useful index to control preventive maintenance activity within a maintenance organization and is defined by

$$I_4 = \frac{\text{TTPM}}{\text{TTEM}} \qquad (2.4)$$

where

I_4 = index parameter,

TTPM = total time spent in performing preventive maintenance,

TTEM = total time spent for the entire maintenance function.

As per the past experience, the value of I_4 should be kept within 20 and 40% limits.

Index V

This index can be used to measure the accuracy of the maintenance budget plan and is expressed by

$$I_5 = \frac{\text{TAMC}}{\text{TBMC}} \qquad (2.5)$$

where

I_5 = index parameter,

TAMC = total actual maintenance cost,

TBMC = total budgeted maintenance cost.

In this case, large variances indicate the need for immediate attention.

•Index VI

This is a useful index for maintenance overhead control and is expressed by

$$I_6 = \frac{\text{TMAC}}{\text{TMC}} \qquad (2.6)$$

where

I_6 = index parameter,

TMAC = total maintenance administration cost.

Index VII

This index is useful in scheduling work and is expressed as follows:

$$I_7 = \frac{\text{PJCED}}{\text{TPJ}} \qquad (2.7)$$

where

I_7 = index parameter,

PJCED = total number of planned jobs completed by established due dates,

TPJ = total number of planned jobs.

The value of I_7 should be high to keep backlogs down.

Index VIII

This index is useful in material control area and is defined by

$$I_8 = \frac{\mathrm{TPJAM}}{\mathrm{TPJ}} \qquad (2.8)$$

where

I_8 = index parameter,

TPJAM = total number of planned jobs awaiting material.

Index IX

This index can be used to measure maintenance effectiveness and is defined by

$$I_9 = \frac{\text{MHEUJ}}{\text{TMMH}} \qquad (2.9)$$

where

I_9 = index parameter,

MHEUJ = man-hours of emergency and unscheduled jobs,

TMMH = total maintenance man-hours worked

Index X

This index can also be used to measure maintenance effectiveness and is expressed by

$$I_{10} = \frac{\text{DTCB}}{\text{TDT}} \qquad (2.10)$$

where

I_{10} = index parameter,

DTCB = downtime caused by breakdowns,

TDT = total downtime.

Index XI

This is an important index used to measure inspection effectiveness and is defined by

$$I_{11} = \frac{\text{NJI}}{\text{TIC}} \qquad (2.11)$$

where

I_{11} = index parameter,

NJI = number of jobs resulting from inspections,

TIC = total number of inspections completed

Index XII

This index relates material and labour costs and is expressed by

$$I_{12} = \frac{\text{TMLC}}{\text{TMMC}} \qquad (2.12)$$

where

I_{12} = index parameter,

TMLC = total maintenance labour cost,

TMMC = total maintenance materials cost

Index XIII

This index relates maintenance cost to manufacturing cost and is defined by

$$I_{13} = \frac{\text{TMC}}{\text{TMFC}} \qquad (2.13)$$

where

I_{13} = index parameter,

TMFC = total manufacturing cost.

Index XIV

This index relates maintenance cost to man-hours worked and is expressed by

$$I_{14} = \frac{\text{TMC}}{\text{TNMW}} \qquad (2.14)$$

where

I_{14} = index parameter,

TNMW = total number of man-hours worked.

Index XV

This is a useful index to monitor progress in cost reduction efforts and is defined by

$$I_{15} = \frac{\text{PMMSJ}}{\text{MCPP}} \qquad (2.15)$$

where

I_{15} = index parameter,

PMMSJ = percentage of maintenance man-hours spent on scheduled jobs,

MCPP = maintenance cost per unit of production.

Questions

1. List at least ten important functions of a maintenance department.
2. What are the advantages and disadvantages of centralized maintenance?
3. Describe a nine-step approach that can be used to manage a maintenance program.
4. Discuss six important maintenance management principles.
5. List ten questions the maintenance managers can use to self-evaluate effectiveness of their overall maintenance management
6. Discuss the following three elements of effective maintenance management:
    a. Maintenance policy
    b. Work order system
    c. Job planning
7. Define two indices that can be used to evaluate overall performance of a maintenance organization

CHAPTER THREE

PREVENTIVE MAINTENANCE

- Preventive maintenance is all actions carried out on a planned, periodic, and specific schedule to keep an item/equipment in stated working condition through the process of checking and reconditioning.

  These actions are precautionary steps undertaken to forestall or lower the probability of failures or an unacceptable level of degradation in later service, rather than correcting them after they occur.

PM may be described as the care and servicing by individuals involved with maintenance to keep equipment/facilities in satisfactory operational state by providing for systematic inspection, detection, and correction of incipient failures either prior to their occurrence or prior to their development into major failure.

Main Objectives of PM

- enhance capital equipment productive life,
- reduce critical equipment breakdowns,
- allow better planning and scheduling of needed maintenance work,
- minimize production losses due to equipment failures,
- promote health and safety of maintenance personnel

PM programs in maintenance organizations end up in failure (i.e., they lose upper management support)
because their cost is either unjustifiable or they take a significant time to show results.

It is emphasized that all PM must be cost effective.

The most important principle to keep continuous management support is: “perform money saver activities!”

Elements of Preventive Maintenance

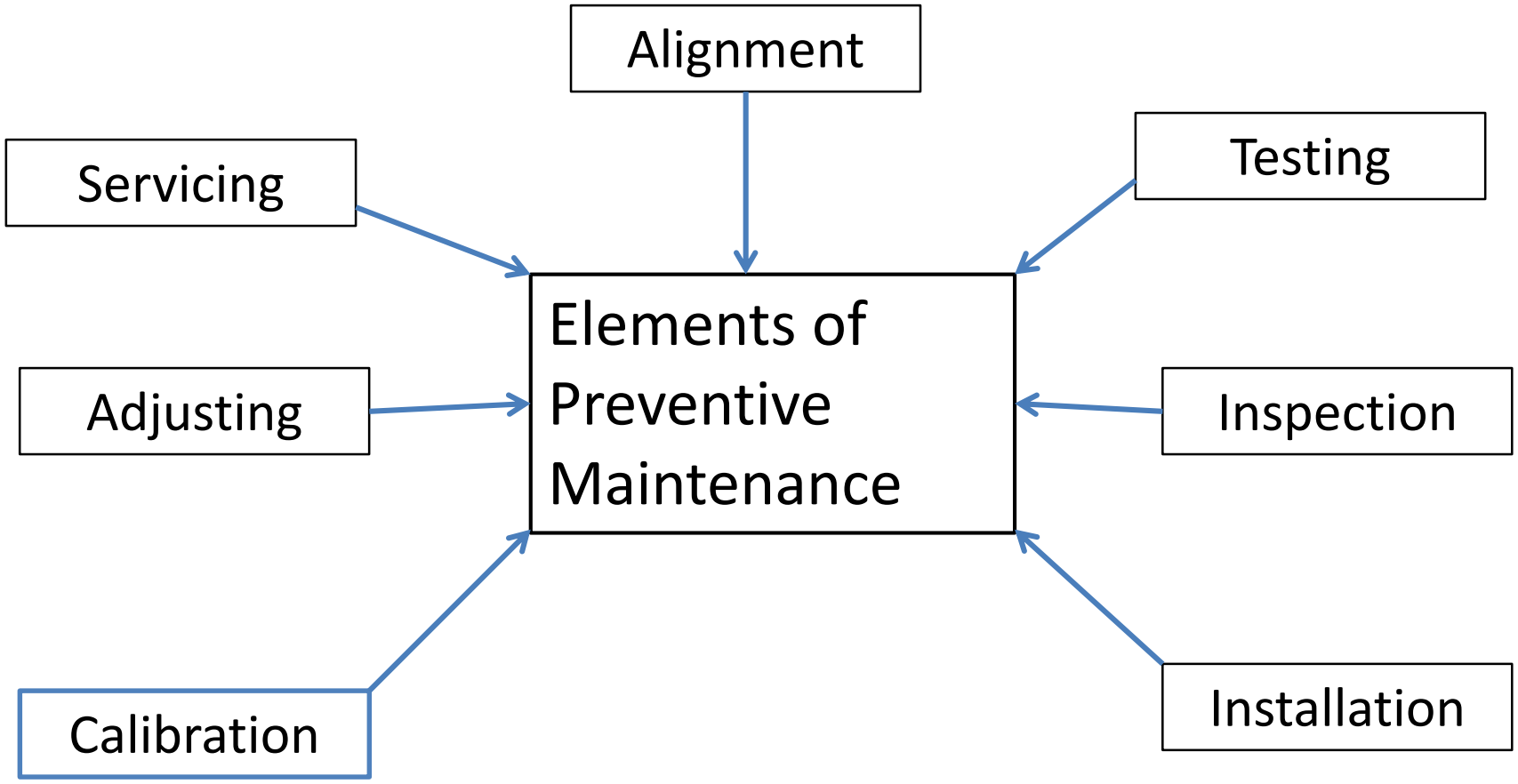


Fig 3.1 Elements of Preventive maintenance

1. **Inspection**: Periodically inspecting materials/items to determine their serviceability by comparing their physical, electrical, mechanical, etc., characteristics (as applicable) to expected standards

2. **Servicing**: Cleaning, lubricating, charging, preservation, etc., of items/materials periodically to prevent the occurrence of incipient failures

3. **Calibration**: Periodically determining the value of characteristics of an item by comparison to a standard; it consists of the comparison of two instruments, one of which is certified standard with known accuracy, to detect and adjust any discrepancy in the accuracy of the material/parameter being compared to the established standard value

4. **Testing:** Periodically testing or checking out to determine serviceability and detect electrical/mechanical-related degradation
5. **Alignment:** Making changes to an item's specified variable elements for the purpose of achieving optimum performance
6. **Adjustment:** Periodically adjusting specified variable elements of material for the purpose of achieving the optimum system performance
7. **Installation:** Periodic replacement of limited-life items or the items experiencing time cycle or wear degradation, to maintain the specified system tolerance

Characteristics of a plant in need of a good preventive maintenance program

- Low equipment use due to failures
- Large volume of scrap and rejects due to unreliable equipment
- Rise in equipment repair costs due to negligence in areas such as regular lubrication, inspection, and replacement of worn items/components
- High idle operator times due to equipment failures
- Reduction in capital equipment expected productive life due to unsatisfactory maintenance

Preventive Maintenance Program Evaluation Questions

- 17 questions for determining the adequacy of a preventive maintenance program within an organization.
- The answer “yes” or “no” to each question is given 5 or 0 points, respectively.
- A “maybe” answer is assigned a score from 1 to 4. A total score of less than 55 points indicates that the preventive maintenance program requires further improvements.

Preventive Maintenance Program Evaluation Questions

	Question	Answer		
		Yes (5 points)	May be (1 -4 points)	No (0 point)
1	**Is the trend in downtime recorded and reported regularly?**			
2	Is there a formal PM program in place?			
3	Are inspectors performing their inspection duties full-time?			
4	Are check sheets controlled to assure 100% compliance?			
5	Are inspection routes developed/scheduled on the basis of work measurement methods?			
6	Are inspection reports randomly checked by supervisor to determine their accuracy?			

Preventive Maintenance Program Evaluation Questions

	Question	Answer		
		Yes (5 points)	May be (1 -4 points)	No (0 point)
7	**What percentage of downtime is due to maintenance?**	(8%)	(8%)	(Unknown)
8	Is the lubrication task performed through the scheduled usage of check sheets?			
9	Does maintenance management receive meaningful downtime reports?			
10	Is one individual responsible for the overall PM?			
11	Were lubrication routes developed and scheduled on the basis of time and method studies?			
12	Is data processing used to schedule and report PM inspections and lubrication?			

Preventive Maintenance Program Evaluation Questions

	Question	Answer		
		Yes (5 points)	May be (1 -4 points)	No (0 point)
13	**Are foreseeable problems, discovered through PM inspections, quickly reported?**			
14	Is PM work highlighted in the cost reporting system to permit routine analysis of PM as a distinct class of expenditure?			
15	Are lubrication requirements examined regularly to minimize the need for different types of lubricants?			
16	Is the analysis of breakdown reports performed to detect failure patterns that can be rectified by adjusting the PM program?			
17	Are plant/building assets examined regularly as an integral part of the formal inspection program?			

Principle or formular for PM program

$$(NB)(ACPBD)(\alpha)>CPMS$$

where

CPMS = total cost of preventive maintenance system,

α = a factor whose value is proposed to be taken as 70%; more specifically, 70% of the total cost of breakdowns,

NB = number of breakdowns,

ACPBD = average cost per breakdown.

PREVENTIVE MAINTENANCE PROGRAM RISKS

- Damage to adjacent equipment during a PM task.
- Damage during the performance of an inspection, repair, adjustment or installation of a replacement part.
- Damage installing material that is defective, incorrect replacement part,
- Damage due to an error in reinstalling equipment into its original status.
- Damage in following wrong reassembling steps.

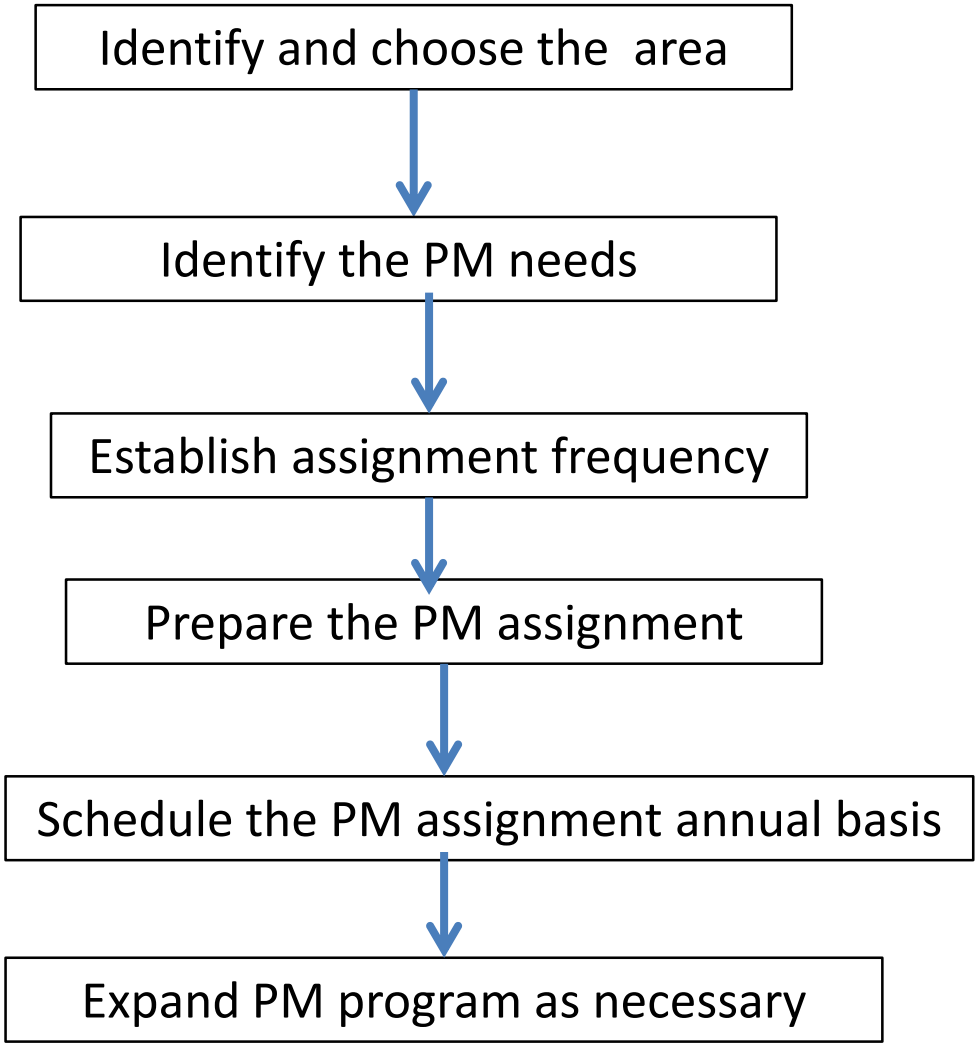
IMPORTANT STEPS FOR ESTABLISHING A PM PROGRAM

- accurate historical records of equipment
- manufacturer’s recommendations
- skilled personnel
- past data from similar equipment
- service manuals
- unique identification of all equipment
- appropriate test instruments and tools
- management support and user cooperation

IMPORTANT STEPS FOR ESTABLISHING A PM PROGRAM

- failure information by problem/cause /action
- consumables and replaceable components/parts
- clearly written instructions with a checklist to be signed off

6 Step for developing PM program



Identify and choose the areas

- Identify and selection of one or two important areas to concentrate the initial PM effort. These areas should be crucial to the success of overall plant operations and may be experiencing a high degree of maintenance actions.

- The main objective of this step is to obtain immediate results in highly visible areas, as well as to win concerned management support.

Identify the PM needs

- Define the PM requirements.
- Establish a schedule of two types of tasks:
  - daily PM inspections
  - periodic PM assignments.
- The daily PM inspections could be conducted by either maintenance or production personnel (e.g. is to check the waste water settleable solids concentration).
- Periodic PM assignments usually are performed by the maintenance workers (e.g. replacing throwaway filters, replacing drive belts, and cleaning steam traps and permanent filters).

Establish assignment frequency

- Establish the frequency of the assignments. This involves reviewing the equipment condition and records.

- the basis for establishing the frequency is the experience of those familiar with the equipment and the recommendations of vendors and engineering.

- vendor recommendations are generally based on the typical usage of items under consideration.

Prepare the PM assignments

Daily and periodic assignments are identified and described in detail, then submitted for approval.

Schedule the PM assignments on annual basis

The defined PM assignments are scheduled on the basis of a twelve-month period.

Expand the PM program as necessary

After the implementation of all PM daily inspections and periodic assignments in the initially selected areas, the PM can be expanded to other areas.

Experience gained from the pilot PM projects is instrumental to expanding the program.

PM ADVANTAGES

- advantages including
  - increase in equipment availability,
  - performed as convenient,
  - balanced workload,
  - reduction in overtime,
  - increase in production revenue,
  - consistency in quality,
  - reduction in need for standby equipment,
  - stimulation in pro action instead of reaction,
  - reduction in parts inventory,
  - improved safety,
  - standardized procedures, times, and costs

PM DISADVANTAGES

disadvantages including:

- exposing equipment to possible damage,
- using a greater number of parts,
- increases in initial costs,
- failures in new parts/components,
- demands more frequent access to equipment/item

CHAPTER FOUR

CORRECTIVE MAINTENANCE

CORRECTIVE MAINTENANCE

- Although every effort is made to make engineering systems as reliable as possible through design, preventive maintenance, and so on, from time to time they do fail .

- Therefore, repair or corrective maintenance is an important component of maintenance activity

CORRECTIVE MAINTENANCE

- may be defined as the remedial action carried out due to failure or deficiencies discovered during preventive maintenance, to repair an equipment/item to its operational state.
- is an unscheduled maintenance action, basically composed of unpredictable maintenance needs that cannot be pre-planned or programmed on the basis of occurrence at a particular time

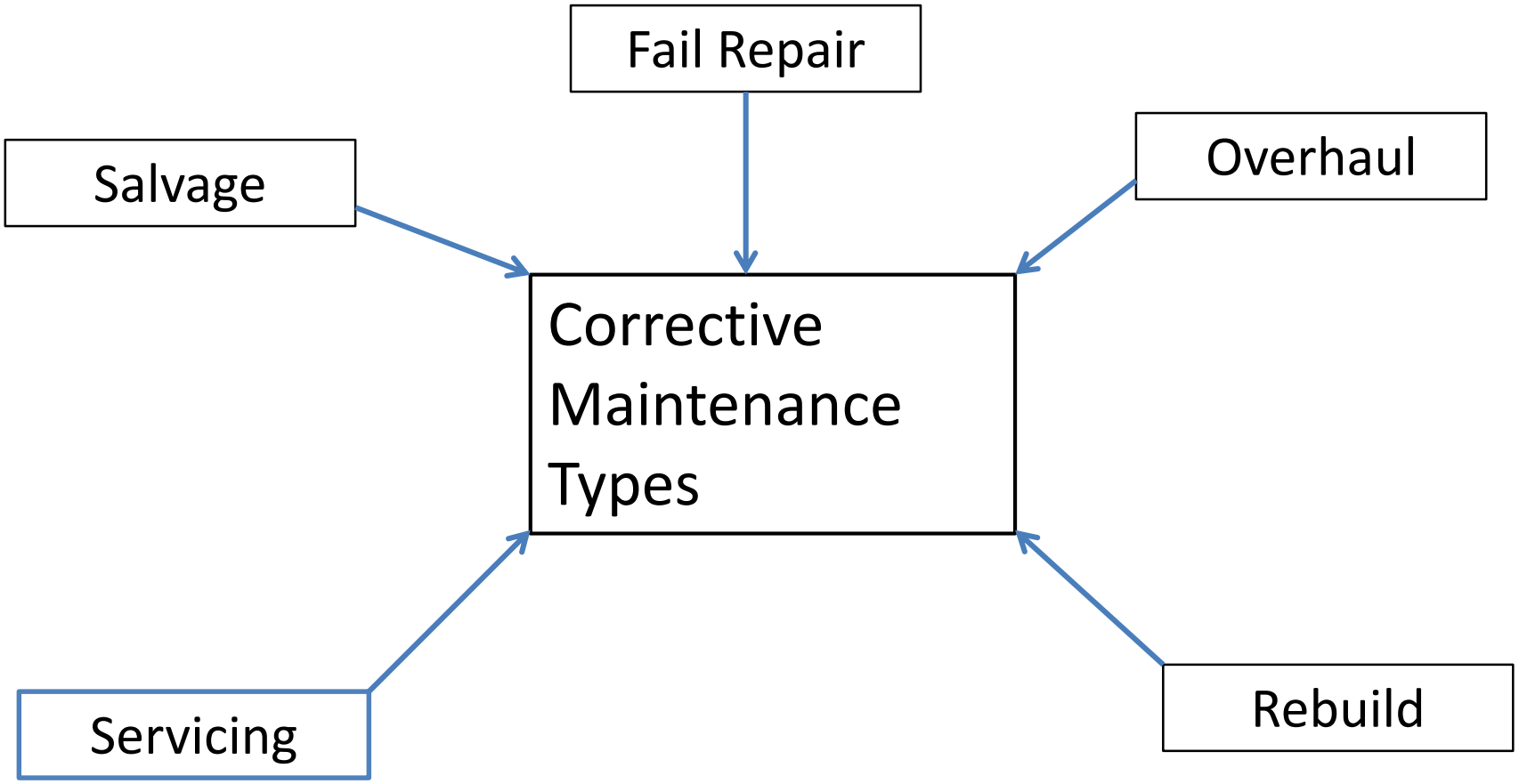
- The action requires urgent attention that must be added, integrated with, or substituted for previously scheduled work items.

- This incorporates compliance with “prompt action” field changes, rectification of deficiencies found during equipment/item operation, and performance of repair actions due to incidents or accidents.

CORRECTIVE MAINTENANCE TYPES

- Corrective maintenance may be classified into five major categories
  - Fail-repair
  - Salvage
  - Rebuild
  - Overhaul
  - Servicing

TYPES OF CORRECTIVE MAINTENANCE



Fail-repair

- The failed item is restored to its operational state

Salvage

- This element of corrective maintenance is concerned with disposal of non-repairable material and use of salvaged material from non-repairable equipment/item in the repair, overhaul, or rebuild programs.

Rebuild

- This is concerned with restoring an item to a standard as close as possible to original state in performance, life expectancy, and appearance. This is achieved through complete disassembly, examination of all components, repair and replacement of worn/unserviceable parts as per original specifications and manufacturing tolerances, and reassembly and testing to original production guidelines.

Overhaul

- Restoring an item to its total serviceable state as per maintenance serviceability standards, using the “inspect and repair only as appropriate” approach.

Servicing

- Servicing may be needed because of the corrective maintenance action, for example, engine repair can lead to crankcase refill, welding on, etc.

  Another example could be that the replacement of an air bottle may require system recharging.

PRE-REQUISITES OF CORRECTIVE MAINTENANCE

- The maintenance program must be able to anticipate maintenance requirement before a breakdown can occur.

- A corrective maintenance program can function well once a comprehensive predictive maintenance program is in place.

- Information from the predictive maintenance program assists in the determination of incipient problems

- Preparation will be made and the equipment can then fall into corrective maintenance program provided all needed replacement program is lined up.
- With assistance of trained maintenance personnel, replacement parts, required tools, proper repair procedure, and adequate time the corrective maintenance will be executed.

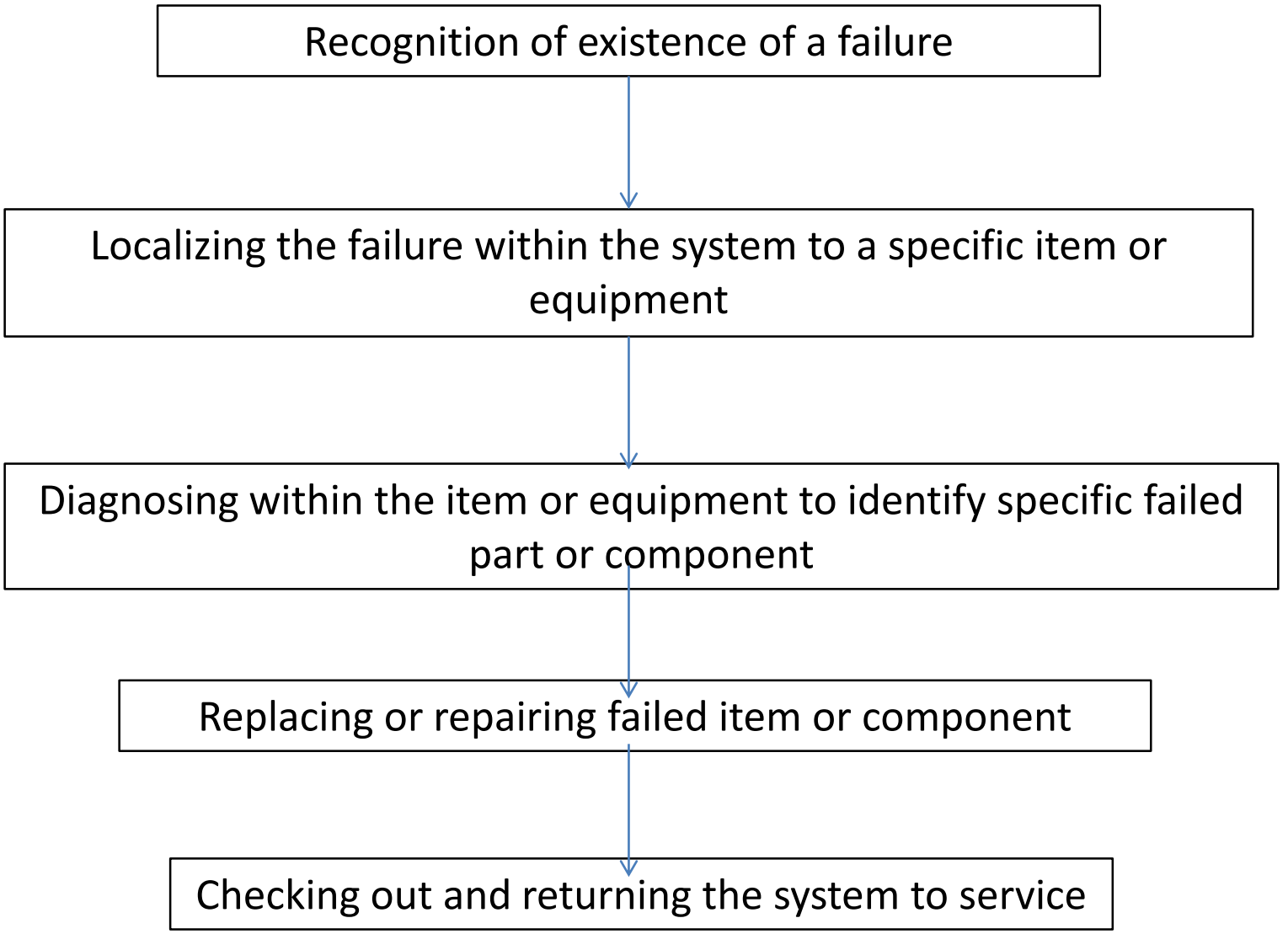
- The serviced part/machine will be verified before usage.

CORRECTIVE MAINTENANCE STEPS

Sequential steps for performing corrective maintenance is:

- Fault recognition
- Localization
- Diagnosis
- Repair
- Checkout

Corrective Maintenance Steps



CORRECTIVE MAINTENANCE DOWNTIME COMPONENT

The major corrective maintenance downtime components are:

- active repair time
- administrative time
- logistic time
- delay time

Logistics delay time LDT

Logistics delay time is the maintenance downtime that is expended as a result of

- waiting for a spare part to become available
- waiting for necessary test equipment to perform maintenance
- waiting for transportation
- waiting to use a facility required for maintenance, and so on.

LDT does not include active maintenance time but does constitute a major element of total maintenance downtime (MDT).

Administrative delay time (ADT)

Administrative delay time refers to that portion of downtime during which maintenance is delayed for reasons of an administrative nature:

- personnel assignment priority,
- labour strike,
- organizational constraint, and so on.

ADT does not include active maintenance time but often constitutes a significant element of total maintenance downtime (MDT)

Active Repair Time

The active repair time is made up of the following subcomponents:

a. Preparation time

b. Fault location time

c. Spare item obtainment time

d. Fault correction time

e. Adjustment and calibration time

f. Checkout time

STRATEGIES FOR REDUCING THE SYSTEM-LEVEL CORRECTIVE MAINTENANCE TIME

Steps for minimizing the time spent on corrective maintenance exercises:-

- Efficiency in fault recognition, location, and isolation
- Effective interchangeability
- Redundancy
- Effective accessibility
- Human factor considerations
- Human factor considerations

Efficiency in fault recognition, location, and isolation

- Past experience indicates that in electronic equipment, fault isolation and location consume the most time within a corrective maintenance activity.
- In the case of mechanical items, often the largest contributor is repair time. Factors such as well designed fault indicators, good maintenance procedures, well-trained maintenance personnel, and an unambiguous fault isolation capability are helpful in lowering corrective maintenance time.

Effective interchangeability

- Good physical and functional interchangeability is useful
- in removing and replacing parts/items,
- reducing maintenance downtime,
- and creating a positive impact on spares and inventory needs

Redundancy

- This is concerned with designing in redundant parts that can be switched in at the moment of need so the equipment/system continues to operate while the faulty part is being repaired.

- In this case the overall Corrective maintenance sequential steps and maintenance workload may not be reduced, but the equipment/system downtime could be impacted significantly.

Effective accessibility

- Often a significant amount of time is spent accessing the failed part.

- Proper attention to accessibility during design can help reduce part accessibility time and, in turn, the corrective maintenance time.

Human factor considerations

- Attention paid to human factors during design in areas such as readability of instructions, size, shape,
- weight of components selection
- placement of dials and indicators,
- size and placement of access, gates, and readability,
- information processing aids can help reduce corrective maintenance time significantly

CHAPTER FIVE

PREDICTIVE MAINTENANCE

PREDICTIVE MAINTENANCE

Predictive maintenance is a management technique that, uses regular evaluation of the actual operating condition of plant equipment, production systems, and plant management functions to optimize total plant operation.

The output of a predictive maintenance program is data

- Until action is taken to resolve the deviations or problems revealed by the program, plant performance cannot be improved.

- Therefore, a management philosophy that is committed to plant improvement must exist before any meaningful benefit can be derived.

- Properly used, predictive maintenance can identify most, if not all, factors that limit effectiveness and efficiency of the total plant.

- One factor that limits the effective management of plants is the lack of timely, factual data that defines operating condition of critical production systems and the effectiveness of critical plant functions, such as purchasing, engineering, and production.

- In a total plant performance mode, predictive technology can be used to accurately measure the effectiveness and efficiency of all plant functions, not just machinery.

- The data generated by regular evaluation can isolate specific limitations in skill levels, inadequate procedures, and poor management methods as well as incipient machine or process system problems.

As a maintenance management tool, predictive maintenance can provide the data required to schedule both preventive and corrective maintenance tasks on an as-needed basis.

Instead of relying on industrial average-life statistics, such as mean-time-to-failure, to schedule maintenance activities, predictive maintenance uses direct monitoring of the operating condition, system efficiency, and other indicators to determine the actual mean-time-to-failure or loss of efficiency for each machine train and system within the plant.

- A predictive maintenance program can minimize unscheduled breakdowns of all mechanical equipment in the plant and ensure that repaired equipment is inacceptable mechanical condition

- To achieve these goals, the predictive maintenance program must correctly identify the root cause of incipient problems.
- Many of the established programs do not meet this fundamental requirement

system vendors

- Precipitated by the claims of predictive maintenance system vendors, many programs are established on simplistic monitoring methods that identify the symptom rather than the real cause of problems

- many of these programs fail because maintenance managers lose confidence in the program's ability to accurately detect incipient problems.

- To be an effective maintenance management tool, it must be combined with a viable maintenance planning function that will use the data to plan and schedule appropriate repairs
- it is dependent on the skill and knowledge of maintenance craftsmen

Predictive maintenance program techniques

- Vibration monitoring
- Broadband Trending
- Narrow band Trending
- Signature Analysis
- Thermography
- Infrared thermometer
- Line Scanners
- Infrared imaging
- Visual inspection
- Ultrasonic monitoring
- Operating dynamics analysis
- Other Techniques

Infrared imaging

- Thermal or infrared imaging provides the means to scan the infrared emissions of complete machines, process, or equipment in a very short time.
- Infrared techniques can be used to detect problems in a variety of plant systems and equipment, including electrical switchgear, gearboxes, electrical substations, transmissions, circuit breaker panels, motors, building envelopes, bearings, steam lines, and process systems that rely on heat retention or transfer.

- 1. Tribology
  - a. Lubricating Oil Analysis
  - b. Spectrographic Analysis
  - c. Wear particle analysis
  - d. Ferrography
- 2. Process parameters
- 3. Electric Motor Analysis
  - a. Insulation resistance
  - b. Other electrical testing

Other Techniques

- Other techniques that can support predictive maintenance include
- acoustic emissions,
- eddy-current,
- magnetic particle,
- residual stress,
- and most of the traditional non-destructive methods

Merits of Predictive

1. Increased component operational life/availability.
2. Allows for pre-emptive corrective actions.
3. Decrease in equipment or process downtime.
4. Decrease in costs for parts and labour.
5. Better product quality.
6. Improved worker and environmental safety.
7. Improved worker moral.
8. Energy savings.
9. Estimated 8% to 12% cost savings over preventive maintenance program.

Demerits of Predictive

- Increased investment in diagnostic equipment.
- Increased investment in staff training.
- Savings potential not readily seen by management

CHAPTER SIX

RELIABILITY CENTERED MAINTENANCE (RCM)

RCM

- This Maintenance technique, involves continuous improvement of Maintenance programs in the most cost-effective terms and in a technically feasible manner. RCM focuses on listing past failures and Maintenance history, and emphasizes the functional importance of system components.

- RCM methodology deals with some key issues not dealt with by other maintenance programs.

•It recognizes that all equipment in a facility is not of equal importance to either the process or facility safety. It recognizes that equipment design and operation differs and that different equipment will have a higher probability to undergo failures from different degradation mechanisms than others.

- It also approaches the structuring of a maintenance program recognizing that a facility does not have unlimited financial and personnel resources and that the use of both need to be prioritized and optimized.

- RCM is a systematic approach to evaluate a facility's equipment and resources to best mate the two and result in a high degree of facility reliability and cost-effectiveness.

- RCM is highly reliant on predictive maintenance but also recognizes that maintenance activities on equipment that is inexpensive and unimportant to facility reliability may best be left to a reactive maintenance approach

How to Initiate Reliability Centered Maintenance

1. Develop a Master equipment list identifying the equipment in your facility.
2. Prioritize the listed components based on importance to process.
3. Assign components into logical groupings.
4. Determine the type and number of maintenance activities required and periodicity using:
   a. Manufacturer technical manuals
   b. Machinery history
   c. Root cause analysis findings - Why did it fail?
   d. Good engineering judgment
5. Assess the size of maintenance staff.

How to Initiate Reliability Centered Maintenance

6. Identify tasks that may be performed by operations maintenance personnel.

7. Analyze equipment failure modes and effects.

8. Identify effective maintenance tasks or mitigation strategies.

MERITS OF RCM

- Can be the most efficient maintenance program.
- Lower costs by eliminating unnecessary maintenance or overhauls.
- Minimize frequency of overhauls.
- Reduced probability of sudden equipment failures.
- Able to focus maintenance activities on critical components.
- Increased component reliability.
- Incorporates root cause analysis.

DEMERITS OF RCM

- Can have significant startup cost, training, equipment, etc.
- Savings potential not readily seen by management

PRO-ACTIVE MAINTENANCE

- Pro-active maintenance can be defined as an extension of predictive maintenance to include the determination of the reasons that machines break down
- Determine what machine components are responsible for the problem.
- Determine the cause of the detected machine fault.
- Take appropriate steps to eliminate the identified cause of the problem. One example of this would be to do a precision alignment on a machine which is prone to bearing failures because of excessive radial loading.

PRO-ACTIVE MAINTENANCE

- Examine and analyze other machines of the same type, and perform the same preventive tasks on them if appropriate.

- Re-design the machine installation or the machine itself to eliminate the possibility of the same problem occurring later. In other words, design the problem out of the machine.

PRO-ACTIVE MAINTENANCE KEY POINTS

- Root cause failure analysis, determines the mechanisms and causes of machine faults.
- The fundamental causes of machine failures can thus be corrected,
- The failure mechanisms can be gradually engineered out of each machinery installation

PRO-ACTIVE MAINTENANCE BENEFITS

- the techniques are natural extensions of those used in a predictive program, and they are easily added to existing programs

- It gradually designs the problems out of the machines over a period of time resulting in,
  - greatly extended machine life,
  - reduced down time,
  - expanded production capacity.